

QuickMenu SaaS Platform

MULTI-TENANT CONTACTLESS DINING & DIRECT NPCI UPI ECOSYSTEM

System Documentation & Architecture Overview | Version 1.0.0

1. EXECUTIVE SUMMARY

QuickMenu is a full-stack, multi-tenant B2B SaaS platform engineered for modern cafés, restaurants, and hospitality businesses. It eliminates physical printed menus and intermediate payment aggregators by providing high-speed digital QR menus, dynamic table ordering, real-time kitchen display systems (KDS), and 0% commission direct bank settlements via the NPCI Unified Payments Interface (UPI).

2. SYSTEM ARCHITECTURE & USER PORTALS

A. Super Admin Master Console

- **Global Tenant Oversight:** Monitor all onboarded cafés, active subscriptions, revenue trends, and operational health.
- **Plan & Pricing Engine:** Configure subscription tiers (Free, Starter, Pro), feature limits, and direct UPI payment IDs.
- **System Controls:** Instant café suspension, automated plan expiry management, and global media asset repository.

B. Café Owner Operational Dashboard

- **Interactive Menu Management:** Create categories and dishes with images, prices, veg/non-veg tags, and instant out-of-stock toggles.
- **Live Real-Time KDS:** Visual order flow across New ! Accepted ! Preparing ! Ready ! Completed with audio alerts.
- **Custom Table QR Generator:** Download high-resolution branded table QR codes with instant redirect URLs.
- **Permanent Revenue Analytics:** Live day/week/month revenue tracking, average order value (AOV), and exportable ledger.
- **Direct Settlement Profile:** Café-specific UPI ID configuration ensuring 100% of customer payments go straight to the owner.

C. Customer Mobile Web App (Zero Install Required)

- **Instant QR Scan Launch:** Customers scan the table QR with any camera or UPI app to access the responsive menu instantly.
- **Real-Time Smart Cart:** Fast search, veg-only filter, customizable spice/prep notes, and instant bill breakdown.
- **Dual Payment Options:** Direct Bank UPI (0% fee deep-linking into GPay, PhonePe, Paytm) or Cash-on-Delivery counter payment.
- **Live Order Tracker:** Real-time order progress tracking with automatic daily sequential token numbers (e.g., #1, #2).
- **Ratings & Reviews:** Integrated post-meal star rating and customer feedback system.

Ø=Ü³ 3. DIRECT NPCI UPI PAYMENT ARCHITECTURE (0% COMMISSION)

Unlike traditional POS systems that charge 2% to 3% gateway transaction fees, QuickMenu uses a direct NPCI UPI protocol. Customer payments bypass intermediaries and land instantly in the café owner's linked bank account.

HOW DIRECT NPCI UPI TRANSACTIONS OPERATE:

1. Session Initiation: Server generates a unique session ID and dynamic intent URL: `upi://pay?pa={upi_id}&am={total}&tr={sessionId}`
2. Customer Action: Customer taps "Pay via UPI" (mobile opens GPay/PhonePe) or scans dynamic QR on desktop/tablet.
3. Direct Bank Settlement: Money transfers instantly bank-to-bank with zero aggregator deductions.
4. Real-Time Webhook & Polling: Backend verifies incoming settlement, marks payment received, and allocates token number.
5. Kitchen Broadcast: Socket.io instantly pings the café dashboard with audio chimes and pushes the order to the KDS.

&™p 4. TECHNOLOGY STACK & SPECIFICATIONS

Frontend Application

- **Framework:** React 19 + Vite for ultra-fast SPA loading and zero-lag rendering.
- **Styling & Design:** Modern Glassmorphism UI, Responsive CSS, TailwindCSS v4, Lucide & React Icons.
- **State & Networking:** Axios API client with JWT interceptors, Socket.io-client for live subscriptions & orders.
- **Hosting Ready:** Optimized for instant deployment on Vercel with SPA routing rewrite rules.

Backend & Cloud Infrastructure

- **Runtime & Server:** Node.js + Express RESTful API architecture with modular routers and validators.
- **Database & Cache:** MongoDB Atlas with Mongoose ODM, indexed schemas, and TTL order retention policies.
- **Real-Time Engine:** Socket.io WebSocket rooms for tenant-isolated café & customer notifications.
- **Security & Protection:** JWT Bearer Authentication, Bcrypt password hashing, express-rate-limit protection.
- **Automated Jobs:** node-cron for daily automated subscription expiry and renewal notification reminders.
- **Cloud Assets:** Cloudinary integration for high-performance café logos and food menu photography.

Ø=þ€ 5. DEPLOYMENT & PRODUCTION SETUP

- **Frontend Hosting:** Vercel / Netlify — Built using "npm run build", static assets served via edge CDN.
- **Backend Hosting:** Railway / Render / VPS — Requires persistent Node.js environment to maintain Socket.io websockets.
- **Environment Variables:** Managed securely via .env (MONGO_URI, JWT_SECRET, CLOUDINARY, SMTP_EMAIL, ADMIN_UPI).

